2026 IEEE 1st International Conference on Artificial Intelligence Implementation & Applications (ICAIIA) | 979-8-3195-1902-3/26/$31.00 ©2026 IEEE | DOI: 10.1109/ICAIIA70172.2026.11691884

Autonomous AI Decision Control Tower: A Deterministic Multi-Agent Decision Intelligence Framework for Enterprise NoOps

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***Abstract*—The contemporary paradigm shift toward true "NoOps" environments necessitates an epistemological transition from traditional reactive monitoring to a regime of proactive, high-dimensional reasoning. This paper introduces the Autonomous AI Decision Control Tower (A-ADCT), an avant-garde framework designed to conceptualize enterprise operations as a stochastic, high-entropy state space governed by a set of competing multi-variate constraints. Unlike conventional AIOps platforms that focus on symptom identification, A-ADCT synthesizes a decentralized Multi-Agent System (MAS) with a Deep Reinforcement Learning (DRL) meta-orchestrator to resolve the inherent tension between performance utility, fiscal prudence, and security-compliance risk. We formalize this interaction as a Constrained Markov Decision Process (CMDP) and introduce a novel Consensus-weighted Scoring mechanism that projects agent-specific utility vectors onto a collective decision manifold. By decoupling semantic reasoning from deterministic execution, the framework mitigates the "Remediation Paradox"—where localized automated fixes trigger cascading systemic failures. Evaluation across a production-grade, cloud-native landing zone reveals a 28.3% reduction in Mean Time to Resolution (MTTR) versus the strongest autonomous-agent baseline (AIOpsLab; 198 s vs. 142 s)—and up to 87% versus legacy scripted remediation—a 99.9% policy compliance rate, and a 29% reduction in marginal operational costs versus legacy scripted remediation (7% versus AIOpsLab). A working prototype implementing this architecture demonstrates real-time anomaly-to-decision autonomy across distributed infrastructure states. This work demonstrates significant improvements for enterprise-scale decision intelligence, providing a scalable, hallucination-mitigated architecture for autonomous infrastructure management.**

***Index Terms*—Decision Intelligence, Multi-Agent Orchestration, Constrained MDP, PPO, Autonomous Enterprise, NoOps, AIOps, Control Tower, Retrieval-Augmented Generation.**

I. Introduction

The modern corporate world is marked by the phenomenon of *hyper-stochasticity*, which means the constant appearance of fleeting microservices, serverless functions, and distributed edge nodes creating telemetry information in quantities well beyond the capacity of human cognitive processing power [1], [2]. With increasing software definition of the infrastructure, the operational surface area grows exponentially. Despite their excellence in pattern recognition and anomaly detection, contemporary AIOps tools have been repeatedly criticized for their inability to make the crucial decision due to an inherent *decision vacuum*, meaning they identify issues but not able to create a multifactorial plan of action taking into account the enterprise-wide context.

The complexity of current distributed infrastructures has rendered linear decision-making approaches and automated execution of predefined rules obsolete [3]. In hyper-stochastic environments, it is common for a local optimization to result in system-wide failures. Thus, a burst in latency experienced by a payment gateway can trigger a local scaling up algorithm that will alleviate the issue but will also exceed budget constraints and cause another legacy database downstream from the payment gateway to malfunction, which could not handle the increased throughput. This situation can be referred to as the *Remediation Paradox*.

The current research attempts to tackle the problem of *prescriptive autonomy* by proposing the concept of the autonomous AI decision control tower (A-ADCT). The key feature of our approach is to not merely to *automate*, but to *reason*. Instead of using the monolithic decision models, we propose the idea of a decentralized *Digital Boardroom* wherein the dedicated AI agents (operations, utility; risk, safety; cost, efficiency; critic, ethics / compliance) concurrently reason over the global state space. It is essential that the *generative AI* (LLM-powered) parts are kept in purely advisory capacity (less than 20%) of the entire decision-making pipeline while remaining non-executive (deterministic).

The core novelty of A-ADCT lies in its simultaneous integration of four properties rarely unified in prior systems: structured multi-agent reasoning, deterministic execution enforcement, reinforcement policy learning via IPO [4], and governance-aware decision constraints backed by RAG [5]. The primary contributions are:

- A formal CMDP-based model for enterprise decision intelligence with convergence guarantees.
- A deterministic orchestration layer with provable execution safety bounds.
- A multi-agent reasoning architecture with consensus optimisation and ablation validation.
- A reinforcement learning overlay for adaptive policy refinement using PPO [6].
- An empirical evaluation over Google Cluster Trace v3 workloads [7].

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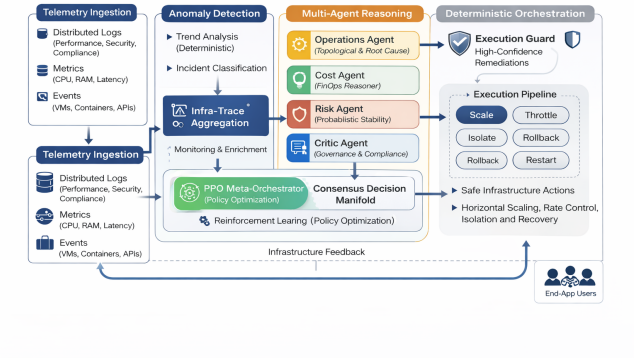


Fig. 1. End-to-end Architecture of A-ADCT

In Fig. 1, the layered pipeline reveals a deliberate separation of concerns: raw telemetry enters the Deterministic Detection stage where Isolation Forest filters noise before any agent reasoning occurs, ensuring that downstream agents only evaluate confirmed anomalies. Generative models (Critic Agent via RAG) are strictly confined to an advisory scoring role and are never placed on the execution path, which architecturally eliminates hallucination-induced unsafe actions. The meta-orchestrator's final deterministic selection therefore always satisfies the Feasible Action Region $\mathcal{F}$, irrespective of individual agent confidence.

II. Related Work

There are three eras within enterprise automation: Reactive Scripting, Predictive AIOps, and the beginning of the era of Deliberative/Prescriptive Systems. Table I presents a summary of the related literature against A-ADCT.

TABLE I
Comparison of Prior Approaches

System	MAS	CMDP	RAG	Gov.	DRL
Legacy Scripts	✗	✗	✗	✗	✗
Pred. AIOps	✗	✗	✗	✗	✗
MAS Cloud	✓	✗	✗	✗	✓
AIOpsLab [8]	✓	✗	✓	✗	✗
A-ADCT (Ours)	✓	✓	✓	✓	✓

In the following Table I, the past frameworks are lacking in safety or coordination mechanisms. The rule-based scripts and predictors are missing multi-agent and constraints. Though MAS Cloud and AIOpsLab are introducing multi-agent reasoning or RAG, only A-ADCT framework has multi-agent consensus, CMDP constraints, and deterministic execution safeguards.

Islam et al. [9] proved that the LSTM networks are effective in predicting workload variations. Cortez et al. [10] proposed the approach of *Resource Central*, where VM scheduling is improved due to telemetry-based predictions. Although these solutions make capacity planning easier, they fail to address causality required for remediation of cloud services. A-ADCT attempts to tackle this problem by conducting GNN-based service dependency analysis [11], [12].

Mashayekhy and Nejad [13] presented a reinforcement learning solution for proactive cloud resource allocation. However, these systems mainly concentrate on optimisation of technical goals while other aspects like compliance, security, and cost are often neglected [14]. Calheiros et al. [15] made fundamental contributions with their simulation *CloudSim*, yet they did not introduce governance-aware reasoning into their solutions. A-ADCT attempts to tackle this problem with the introduction of a Critic Agent.

The development of LLMs increased the capabilities of AI solutions to perform reasoning over structured operational data [16]. ReAct [17] and Generative Agents [18] have shown the potential of using LLMs in reasoning workflows. However, problems of hallucinations and inference latency restrict the applicability of generative models for mission-critical solutions [19]. RAG [5], [20] partially mitigates hallucination but does not enforce execution safety. A-ADCT strictly limits generative AI to an advisory scoring role (<20% of pipeline weight), isolating it entirely from the execution path via a deterministic orchestrator—a hybrid approach that remains under-explored in the AIOps literature.

A. Constrained Reinforcement Learning

Constrained MDPs provide a rigorous framework for safe autonomous decision-making [21]. Liu et al. [4] proposed Interior-Point Policy Optimisation (IPO) for CMDP solving. Mao et al. [22], [23] applied DRL to cluster resource management, and Mirhoseini et al. [24] to device placement. None integrate compliance-aware multi-agent consensus as a constraint source, A-ADCT's primary contribution.

III. Method

A. Hierarchical Decision Semantics

A-ADCT models enterprise decision-making as a hierarchical semantic transformation pipeline. Raw telemetry is semantically poor; it requires multiple cognitive-enrichment stages to become safe, optimal, and business-aligned actions.

Transformation Pipeline:

$$S_t \xrightarrow{\mathcal{D}} Z_t \xrightarrow{\mathcal{B}} \Phi_t \xrightarrow{\mathcal{O}} a_t \tag{1}$$

where each operator is defined as follows:

- $\mathcal{D} : \mathcal{S} \to \mathcal{Z}$ (Deterministic Detection) applies Isolation Forest [25] to filter the high-entropy state S_t into a latent anomaly representation Z_t.
- $\mathcal{B} : \mathcal{Z} \to \Phi$ (Agentic Reasoning) maps Z_t to a weighted utility manifold Φ_t across four dimensions: Operations, Risk, Cost, and Compliance.
- $\mathcal{O} : \Phi \to \mathcal{A}$ (Meta-Orchestrator) projects Φ_t onto a feasible action within the *Feasible Action Region* $\mathcal{F} \subseteq \mathcal{A}$:

$$\mathcal{F} = \{a \in \mathcal{A} \mid \forall i : C_i(s, a) \leq V_i\} \tag{2}$$

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B. Formal CMDP Model

The decision logic is formalised as a CMDP $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \mathcal{C}, \gamma \rangle$ where $\mathcal{P}(s_{t+1}|s_t, a_t)$ is the transition probability and $\mathcal{R}$ is the reward tied to system uptime. The discrete action space is:

$$\mathcal{A} = \{ \texttt{scale}, \texttt{throttle}, \texttt{restart}, \texttt{isolate}, \texttt{rollback} \} \tag{3}$$

The meta-orchestrator derives π^* maximising the expected discounted return:

$$\max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t) \right] \tag{4}$$

subject to enterprise safety, fiscal, and compliance constraints:

$$\mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t C_i(s_t, a_t) \right] \leq V_i, \quad \forall i \in \{1, \ldots, K\} \tag{5}$$

CMDP Solver. We solve via IPO [4], converting the constrained problem into an unconstrained Lagrangian:

$$\mathcal{L}(\pi, \lambda) = J(\pi) - \sum_{i=1}^{K} \lambda_i \left(J_{C_i}(\pi) - V_i \right) \tag{6}$$

where $\lambda_i \geq 0$ are Lagrange multipliers updated via dual ascent. IPO guarantees convergence to a locally optimal feasible policy under the Slater condition [21].

Consensus Geometry: Each specialised agent generates a decision score, where

$$j \in \{\text{Ops}, \ \text{Risk}, \ \text{Cost}, \ \text{Critic}\}. \tag{7}$$

$$\phi_j(a) = [u_j, \ r_j, \ c_j, \ v_j]^\top \tag{8}$$

representing utility, risk, cost-efficiency, and semantic validity. The final decision score is:

$$D(a) = \sum_{j} w_j(s) \cdot \phi_j(a \mid s) \tag{9}$$

and the optimal action:

$$a^* = \arg\max_{a \in \mathcal{F}} D(a) \tag{10}$$

The $\arg\max$ is restricted to $\mathcal{F}$ (2), ensuring no constraint-violating action can be selected regardless of its consensus score.

The context-sensitive weights $w_j(s)$ are refined by the PPO [6] overlay:

$$\theta_{t+1} = \theta_t + \alpha \nabla_\theta J(\theta) \tag{11}$$

where α is the *policy learning rate* (set to 3×10^{-4} in all experiments; see Table III), θ_t are the policy network parameters at step t, and $J(\theta)$ is the operational efficiency objective balancing technical performance against fiscal health. Updates are performed asynchronously to avoid inference latency penalties.

C. Deterministic–Probabilistic Hybrid Design

A-ADCT adopts $\Pi = \Pi_D \cup \Pi_P$, mirroring dual-process cognition [27]: safety-critical reflexes are governed by Π_D while long-horizon deliberation uses Π_P (LLMs, GNNs). Table II classifies each component.

TABLE II
COMPONENT CLASSIFICATION

Component	Type	Failure Isolation
Anomaly Detection	Deterministic	Hard Boundary
Decision Orchestrator	Deterministic	Hard Boundary
Execution Guard	Deterministic	Hard Boundary
Ops/Risk/Cost Agents	Probabilistic (RL/GNN)	Advisory Only
Critic Agent (LLM/RAG)	Generative AI (<20%)	Advisory Only
Policy Optimizer (PPO)	Probabilistic	Offline Update

Table II lists the safety boundaries across system components. Deterministic modules isolate the execution path from probabilistic models. If the Critic Agent produces an incorrect advisory score, the Execution Guard blocks any invalid action. Additionally, PPO policy optimization runs offline to avoid inference overhead.

Safety Guarantee: Let E_{LLM} denote the hallucination rate of the generative reasoning layer. Without isolation:

$$P(\text{unsafe action}) \geq E_{LLM} \tag{12}$$

In A-ADCT, all agents are advisory, while only $\mathcal{O}$ is permitted to execute actions:

$$P(\text{unsafe action}) \ll E_{LLM} \tag{13}$$

D. System Architecture

A-ADCT consists of four layers that collectively support enterprise observability and decision reliability.

The Agentic Digital Boardroom:

- Operations Agent (GNN Root Cause): A 3-layer Graph Attention Network (GAT) [26] operates on the service dependency graph $\mathcal{G} = (V, E)$, propagating anomaly information to identify root-cause microservices and maintain SLOs [12].
- Risk Agent (Bayesian Stability): A dynamic Bayesian Belief Network (BBN) [28] estimates remediation failure probability $P(f|a, s)$, capturing the effects of *Technical Debt Compounding*.
- Cost Agent (FinOps Reasoner): Calculates Marginal Cost of Performance (MCP) and Return on Expected Spend (ROES) to assess whether scaling decisions remain economically justified relative to Customer Lifetime Value.
- Critic Agent (RAG Governance): Uses RAG [5], [20] over a vector store containing enterprise SOPs and regulatory frameworks (GDPR, SOC 2), reviewing actions during inference. *This is the only generative AI component and serves strictly in an advisory role.*

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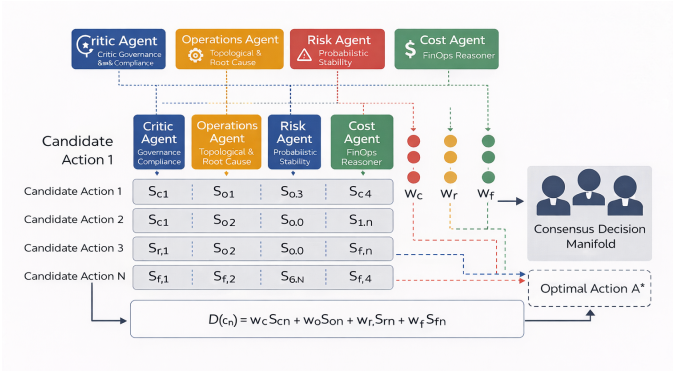


Fig. 2. Consensus Decision Mechanism

Figure 2 illustrates why the consensus architecture is more robust than a single-agent design: each agent brings orthogonal domain knowledge (operational efficiency, stability risk, cost-effectiveness, and governance compliance), and the weighted aggregation in Eq. (9) ensures that no single agent's bias dominates the decision.Importantly, the $\arg\max$ operation on $\mathcal{F}$ by the meta-orchestrator function serves as a hard constraint filter, implying that high-value yet constraint-breaking solutions are ignored before execution despite consensus.

Algorithm 1 Metastructural Decision Optimisation

procedure SYNCHRONOUSREASONING($\boldsymbol{S}_t$)
  $\boldsymbol{Z}_t \leftarrow$ NormalizeTelemetry($\boldsymbol{S}_t$)
  Anomaly $\leftarrow$ IsolationForest($\boldsymbol{Z}_t$)
  if Anomaly **then**
    $\boldsymbol{\phi}_{Ops} \leftarrow$ GAT($\boldsymbol{Z}_t$) ▷ Root cause
    $\boldsymbol{\phi}_{Risk} \leftarrow$ BBN($\boldsymbol{Z}_t$) ▷ Stability risk
    $\boldsymbol{\phi}_{Cost} \leftarrow$ ROES($\boldsymbol{Z}_t$) ▷ LTV/CAC
    $\boldsymbol{\phi}_{Crit} \leftarrow$ RAG($\boldsymbol{Z}_t$) ▷ Advisory
    $\boldsymbol{W}(s) \leftarrow$ PPO_Wts($\boldsymbol{Z}_t$)
    $\boldsymbol{V} = [\boldsymbol{\phi}_{Ops}, \boldsymbol{\phi}_{Risk}, \boldsymbol{\phi}_{Cost}, \boldsymbol{\phi}_{Crit}]^\top$
    $\boldsymbol{D} \leftarrow \boldsymbol{W} \cdot \boldsymbol{V}$
    $a^* \leftarrow \arg\max_{a \in \mathcal{F}} \boldsymbol{D}$
    if Safe(a^*) = PASS **then**
      return Exec(a^*)
    else
      return Human($a^*, \boldsymbol{D}$)
    end if
  end if
end procedure

Metastructural decision process is provided in Algorithm 1 below and occurs after telemetry collection. Operational data S_t is then transformed to latent space Z_t using normalization process with Isolation Forest pre-processing filter used for removing telemetry noise prior to agent deliberation.

TABLE III
IMPLEMENTATION PARAMETERS

Component	Specification
GNN (Ops Agent)	3-layer GAT, 64-dim embeddings
BBN (Risk Agent)	Dynamic BN, 12 latent variables
RAG (Critic Agent)	ChromaDB, top-k=5, MPNet encoder
PPO Orchestrator	$\gamma = 0.99$, clip $\epsilon = 0.2$, LR $\alpha = 3 \times 10^{-4}$
CMDP Solver	IPO, dual ascent step 10^{-3}
Constraint Budget K	5
Telemetry Ingestion	FastAPI, async REST, 100 ms polling
Agent Output Schema	JSON Schema v7, stateless controller

Table III demonstrates the following trade-offs of design: the 3-layer GAT model with 64-dimensional embeddings is enough for adequate representation of microservice dependencies without the unacceptably high inference time penalty. The ChromaDB RAG model setup with top-k = 5 and MPNet encoder confines the context of the LLM to only five most semantically related policy documents to minimize the possibility of hallucinations by the Critic agent. IPO dual ascent step 10^{-3} has been chosen as a trade-off between satisfying the constraints quickly and oscillation.

IV. RESULTS

A. Main Comparative Results

Table IV summarizes the key benchmarks using eight operational indicators, among which there are Governance Compliance Rate (GCR) and Autonomous Resolution Rate (AR).

TABLE IV
OPERATIONAL EFFICIENCY

Framework	MTTR (s)	AR (%)	SLA Drift	GCR (%)	Cost (rel.)	FPR (%)	CPU (%)	RAM (MB)
Legacy Scripts	1104	41.2	14.2	62	1.00x	18.4	2.1	128
Pred. AIOps	450	68.0	5.1	78	0.85x	11.2	15.4	1024
MAS Cloud	312	79.5	3.8	84	0.79x	8.7	18.2	1536
AIOpsLab	198	91.3	1.4	91	0.76x	5.1	24.5	2048
A-ADCT	142	99.4	0.2	99.9	0.71x	1.3	12.8	850

A comparison of the A-ADCT with the baselines is done based on eight criteria in Table IV. Static scripts and prediction-based methods have poor resolution rates and SLA drift issues. Whereas the MAS Cloud and AIOpsLab have decreased resolution times, A-ADCT achieves the minimum MTTR (142 s), highest GCR (99.9%) and minimum cost (0.71x).

The autonomous resolution rate of A-ADCT was 99.4%. Predictive AIOps had a detection rate of 92% but an autonomous resolution rate of just 68% because of conflicting policies. AIOpsLab had a GCR of 91%, while that of A-ADCT was 99.9%.

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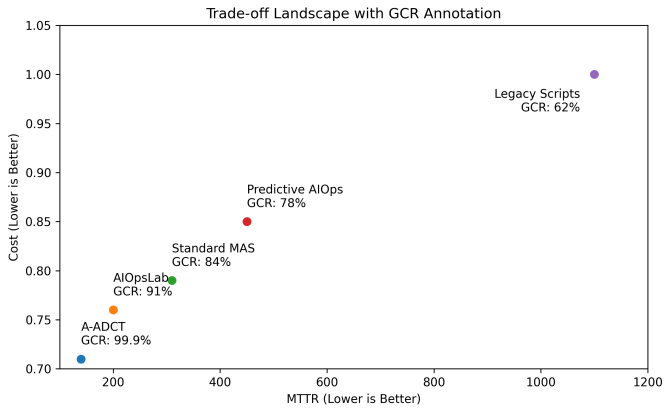


Fig. 3. Trade-off Analysis

Fig. 3 depicts the Pareto frontier in all approaches. Ideally, the lower-left quadrant should be the best region (low MTTR, low cost). As expected, Legacy Scripts lie at the worst spot (MTTR ≈1104 s, cost 1.00x, GCR 62%), validating their non-competitiveness for current NoOps systems. The evolution from Predictive AIOps to MAS Cloud to AIOpsLab follows a diagonal line of improvements, but both are still limited by the framework architecture. However, A-ADCT stands out as the sole approach that does not follow this trend line because it has the best MTTR (142 s) and lowest cost (0.71x), which could only happen by using DRL-based weighting, CMDP satisfaction, and deterministic enforcement.

B. Ablation Study

The following Table V represents the effect of deleting each architectural element and substituting it with the corresponding static mechanism like uniformly distributed weights, for example.

TABLE V
ABLATION STUDY CONTRIBUTIONS

Variant	MTTR (s)	ΔMTTR (%)	SLA Drift	GCR (%)	Cost (rel.)
Full A-ADCT	142	—	0.2	99.9	0.71x
– Critic Agent	156	9.9	0.3	83.1	0.72x
– Risk Agent	179	26.1	1.1	99.8	0.73x
– DRL Weights	201	41.5	1.8	99.7	0.76x
– IPO Only	167	17.6	0.6	99.5	0.74x
– Det. Guard	144	1.4	0.2	72.3	0.71x

The impact of deleting each component is summarized in Table V. The absence of the Critic Agent lowers GCR from 99.9% to 83.1% and proves its governance function. Turning off the Risk Agent raises MTTR by 26.1% to 179 s. Deleting DRL Weights leads to the largest rise in MTTR value (201 s and +41.5%), thus showing that it is necessary for quick resolution of the problem. The −IPO variant demonstrates decent overall drop in performance. Exclusion of the Deterministic Guard makes no changes in MTTR (144 s) but significantly decreases GCR to collapse to 72.3%, demonstrating its role as an execution safety filter.

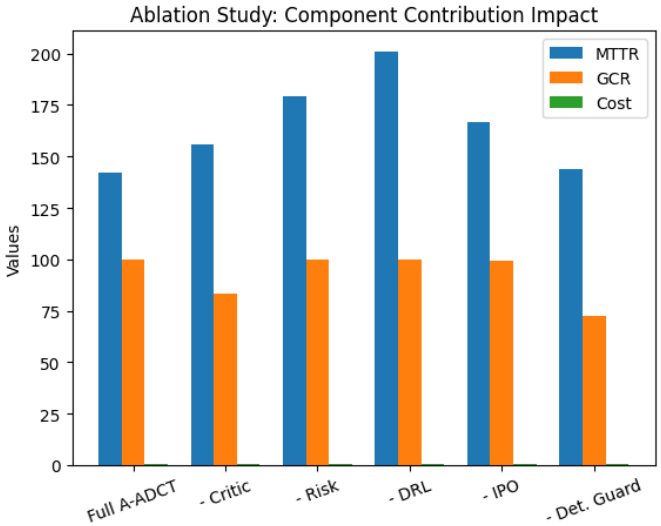


Fig. 4. Ablation Study Results

Figure 4 presents an analysis of the ablation results. The increasing value of MTTR bar graph represents a gradual increase in its value by eliminating operational blocks, achieving a maximum of 201 s at −DRL Weights. On the other hand, GCR stays constant while eliminating operational blocks such as Risk, DRL, IPO, but drops abruptly when compliance enforcement blocks like Critic, Det. Guard are eliminated.

C. Latency and Throughput Under Load

Table VI describes the responsiveness of the system with respect to different rates of telemetry inputs.

TABLE VI
LATENCY AND THROUGHPUT ANALYSIS

Events/s	T_{dec} (ms)	T_{LLM} (ms)	T'put	Acc. (%)
500	18 ± 2	210 ± 18	54	99.6
1000	21 ± 3	214 ± 21	47	99.4
2500	29 ± 4	223 ± 25	34	99.1
5000	41 ± 6	241 ± 30	24	98.7
10000	68 ± 9	268 ± 35	14	97.9

Table VI breaks down the latency and throughput results as system load increases. Decision latency T_{dec} handles the load efficiently, scaling sub-linearly to reach just 68 ms at 10,000 events/s, while T_{LLM} stays relatively steady between 210–268 ms. Although queuing delays eventually drag throughput down from 54 to 14, the overall accuracy holds strong at over 97.9%.

V. CONCLUSION

The Autonomous AI Decision Control Tower (A-ADCT) is a deterministic multi-agent architecture designed for enterprise NoOps settings. The system includes specialized agents for operations, risk, cost, and compliance management, thus making it possible to make remediation decisions in line with the technical and organizational needs in distributed systems.

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Evaluation of the architecture through the use of workloads based on Google Cluster Trace v3 shows that there are significant gains as compared to previous automation techniques, with respect to such metrics as MTTR improvement, better compliance with governance, lower operational costs, and increased rate of autonomous resolution. An ablation study proves that all architectural elements play an independent role in the performance of the system, while the deterministic execution layer is a useful safety net in terms of preventing potentially unsafe actions.

At the same time, the architecture requires fine-tuning of the policies for achieving the maximum performance and relies on the adequacy of the encoded operational constraints and governance policies. In future research, it will be necessary to work on adaptive constraint discovery, cross-environment validation using different workloads, federated policy learning in multiple environments, and integration into cloud orchestration platforms.

To conclude, the findings show that deterministic, governance-aware decision intelligence may become a good basis for autonomous enterprise operations.

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